Industrial Defect Detection

Evaluator Q&A; and viva preparation guide

Use this as a speaking guide, not a script to memorize word for word. Lead with the evidence, state what the notebook measured, and label assumptions or future work clearly.

30-second project summary

“I built a six-class image classifier for hot-rolled steel surface defects using the NEU Surface Defect Database. I compared a small CNN trained from scratch with ResNet-50 and EfficientNet-B0 using the same stratified train/validation/test split. ResNet-50 and EfficientNet-B0 both reached 99.63% accuracy and macro F1 on the 270-image test set. EfficientNet-B0 is the more parameter-efficient choice in this benchmark: it matches ResNet-50 with about one-sixth the parameters. The result is promising, but it needs testing on separate production data before claiming factory readiness.”

Key figures

Dataset	1,800 grayscale images; 200 × 200 pixels; six defect classes; 300 images/class.
Split	Stratified 70/15/15: 1,260 train; 270 validation; 270 test; random_state=42.
Input	Grayscale repeated across 3 channels; resized 224 × 224; ImageNet mean/std normalization.
Best benchmark models	ResNet-50 and EfficientNet-B0 tie at 99.63% accuracy and macro F1.
Model size	Baseline 0.36M; ResNet-50 24.03M; EfficientNet-B0 4.02M parameters.
Error	1 of 270 test images: a scratch predicted as inclusion.

Suggested answer pattern

Answer in this order: choice (what you did), reason (why), evidence (what the notebook measured), then boundary (what the evidence does not establish).

1. Problem, data, and model choice

What problem does the system solve?

It classifies a surface image into one of six known defect classes. It predicts one label for the whole image; it does not locate or segment the defect.

Why use deep learning for this task?

The classes differ through visual structures such as lines, textures, and local intensity patterns. CNNs can learn hierarchical representations from the images rather than relying only on manually specified descriptors.

Why transfer learning?

Only 1,260 images are in the training partition. ImageNet pretraining supplies a useful visual initialization that can be adapted to the steel domain.

Why compare a baseline CNN, ResNet-50, and EfficientNet-B0?

The baseline measures a scratch-trained compact CNN. The other two test transfer learning with different architectures. The comparison checks both performance and parameter count.

Why is EfficientNet-B0 the winner if it ties ResNet-50?

They tie on reported test accuracy and macro F1. EfficientNet-B0 uses 4.02M parameters versus 24.03M, so it is more parameter-efficient here. The benchmark does not prove it is universally better.

Why not choose the smallest baseline?

The 0.36M-parameter CNN scored 68.15% accuracy, well below the transfer-learning models. Its smaller size did not compensate for the performance gap in this experiment.

Why not use HOG, LBP, or an SVM?

Those would be reasonable additional baselines, but the notebook does not evaluate them. I can only compare the three CNNs that were tested.

What is the dataset source?

The notebook identifies the NEU Surface Defect Database and cites Song and Yan (2013), on hot-rolled steel strip surface defects.

2. Preprocessing and representations

Why convert grayscale to RGB?

The pretrained backbones expect three-channel input. Repeating the same grayscale image across three channels matches that input shape without adding color information.

Why resize to 224 × 224?

It gives every model a consistent image size and matches the common input setup for these pretrained architectures.

Why normalize with ImageNet statistics?

The backbones use ImageNet-pretrained weights. This normalization follows their standard input convention; it does not imply that steel images resemble natural images.

What augmentation did you use?

Training used horizontal and vertical flips, 90-degree rotations, affine scale/shift/rotation, CLAHE, brightness/contrast changes, and Gaussian noise. Validation and test used resize and normalization only.

Why augment?

Augmentation creates plausible variation from a small training set and may reduce sensitivity to orientation, position, contrast, and noise.

Why use CLAHE?

It can increase local contrast and reveal faint surface patterns. The notebook demonstrates the visual effect, but does not run an ablation proving CLAHE raised test accuracy.

Could the augmentations hurt?

Yes. A transformation is useful only if it preserves the label and reflects plausible camera or production variation. Orientation changes in particular should be checked against real operating conditions.

Which visual features appeared useful?

Grad-CAM overlays suggest the model attends to defect regions. The notebook interprets cues as line patterns for scratches, fine texture for crazing, and intensity anomalies for inclusions.

Does Grad-CAM prove those features caused the predictions?

No. It is a qualitative, post-hoc visualization of activation associated with a prediction. It is not a causal explanation or a rigorous feature ranking.

Were handcrafted features used?

No class-specific handcrafted descriptors were used. The CNN learns its representations; CLAHE and augmentation are image-preprocessing choices.

3. Training and reproducibility

How were the models trained?

The notebook loads saved checkpoints for evaluation. The project training script uses cross-entropy loss, AdamW, weight decay, and a cosine learning-rate scheduler.

How does ResNet transfer learning work in the project?

The training script first freezes the backbone and trains the classifier head, then unfreezes the upper ResNet layers and fine-tunes them at lower learning rates.

How was EfficientNet fine-tuned?

The project script initializes EfficientNet-B0 with pretrained weights, replaces the classifier for six classes, and optimizes the model parameters.

How was the checkpoint selected?

The training script saves the checkpoint with the best validation accuracy. The test set is intended for the final comparison.

What are the training settings?

The project script defaults to batch size 32 and 12 epochs. It also sets model-specific learning rates and uses cosine annealing. However, the saved curves show different epoch counts, so verify run logs before stating the exact settings used for each checkpoint.

How do you know the split is comparable across models?

The notebook evaluates all three models using the same test loader and stratified split. The data split uses random_state=42.

What about random seeds and repeatability?

The split seed is specified, but the notebook does not report complete seeds for model initialization, augmentation, or training. Repeated seeded runs would provide a better estimate of variance.

4. Metrics and error analysis

How was performance evaluated?

On the 270-image test set using accuracy, macro F1, macro precision/recall, a per-class classification report, and confusion matrices.

Why report macro F1?

It averages F1 across classes with equal weight, which helps reveal weak classes even when overall accuracy is high. The benchmark is balanced, but per-class quality still matters.

What does 99.63% accuracy mean?

The model classified 269 out of 270 test images correctly. One image was incorrect.

What was the error?

A true scratch was predicted as an inclusion. The predicted inclusion probability was 86.0%; the true scratch probability was 9.3%.

Which classes had weaker results?

Scratches had 97.78% recall. Inclusion had 100% recall but 97.83% precision because the scratch error was predicted as inclusion. The other four classes had 100% precision and recall in this test output.

Why can accuracy and macro F1 both be 99.63%?

There was only one error and the test set has equal support per class. The scores are therefore very close, although the metrics measure different things.

Is the model statistically proven to be better than ResNet-50?

No. Their reported point estimates are tied, there are no repeated runs or confidence intervals, and a 270-image test set is limited. The parameter count supports efficiency, not statistical superiority.

How do you interpret precision versus recall for inclusion?

Inclusion recall is 100% because all 45 actual inclusions were found. Precision is 97.83% because 46 images were predicted as inclusion, including the one scratch error.

5. Limitations, risks, and improvements

What are the main limitations?

The dataset is small, balanced, and single-source. Evaluation uses one fixed holdout and has no separate mill, camera, or production-line test.

Could the split have leakage?

The notebook splits individual images but does not describe grouping by acquisition sequence, production lot, or source. I cannot claim leakage occurred; a group-based split and near-duplicate audit would strengthen the evidence.

Are the demo images truly unseen?

That is not verified. The demo picks images from the same dataset by filename order and does not check whether they were used in train, validation, or test. I would call it an inference demo, not independent validation.

Can the model handle a clean surface or a new defect?

That is not demonstrated. It is a closed-set classifier with six defect labels and will choose one of them. It needs an unknown/normal rejection strategy if those inputs can occur.

Does the system locate defects?

No. It classifies the whole image. Grad-CAM is not a validated segmentation or localization output.

Are softmax confidences reliable?

Not necessarily. The notebook reports softmax probabilities but does not evaluate calibration. Confidence should be calibrated and tested before it drives an automated decision.

What would you improve first?

Run an external test on separate production data. Then use group-aware splits, repeated runs or cross-validation, confidence intervals, and more difficult examples. For deployment, assess calibration and a human-review threshold.

What would deployment require?

External validation on the target line, realistic end-to-end timing, handling of unknown or uncertain inputs, and monitoring for drift. Add localization if operators need defect position.

Why is the test result not enough for production?

The test set measures performance on one small partition of NEU. Production may differ in lighting, camera, material, defect severity, and defect frequencies.

Quick-fire follow-up questions

These are short counter-questions evaluators often use to check whether you understand the details behind the headline score.

Name the six classes.

Crazing, inclusion, patches, pitted surface, rolled-in scale, and scratches.

How many images are in the test set per class?

45 per class, 270 total.

What does “stratified” mean here?

Each split preserves approximately the same class proportions as the full dataset.

Why not oversample?

The supplied benchmark has 300 images for every class.

Were validation and test augmented?

No random augmentation; resize and ImageNet normalization only.

What does CLAHE stand for?

Contrast Limited Adaptive Histogram Equalization.

What does the affine augmentation change?

It randomly varies scale, translation, and rotation within configured ranges.

What is the loss function?

Cross-entropy loss in the project training script.

Which optimizer is used?

AdamW, with learning rates set by model and training stage.

What is weight decay for?

It regularizes weights and can reduce overfitting; the script sets it to 1e-4.

What is the batch size default?

32 in the project training script.

What scheduler is used?

CosineAnnealingLR.

How many parameters does EfficientNet-B0 have here?

4.02 million.

How many fewer parameters than ResNet-50?

About six times fewer: 24.03M divided by 4.02M is about 6.

What is a false positive in this result?

The scratch image predicted as inclusion is a false positive for the inclusion class.

What is a false negative in this result?

That same scratch image is a false negative for the scratches class.

Why does inclusion precision fall below 100%?

One non-inclusion image, a scratch, was predicted as inclusion.

Is there a “normal” class?

No normal class appears in this six-class evaluation.

What is the input image’s original color format?

Grayscale JPEG images, according to the notebook’s dataset description.

What device did the notebook report?

An NVIDIA GeForce RTX 4050 Laptop GPU.

Is the reported latency guaranteed on a factory computer?

No. It is a run-specific measurement and should be benchmarked on target hardware.

Did the notebook compare inference costs end to end?

No. Its timing is model forward-pass timing, not a complete camera-to-decision deployment benchmark.

Can the model say “I do not know”?

Not in the evaluated setup; it always chooses one of the six classes.

What is one explainability limitation?

Grad-CAM gives a coarse visual cue and does not prove causal reasoning.

What is the first external validation you would run?

Hold out images from a separate mill, camera, or production lot and evaluate class-wise metrics.

Why use a group split?

It reduces the chance that related images or capture conditions appear across train and test partitions.

What is an ablation study?

A controlled comparison that removes or changes one component, such as CLAHE, to measure its contribution.

What does calibration measure?

Whether predicted confidence corresponds to observed correctness frequency.

Why might the model fail under new lighting?

Lighting and contrast can shift image appearance away from the training distribution.

Would higher resolution always help?

Not necessarily; it may preserve fine defects but costs more computation. Compare resolutions on external validation data.

Important consistency checks

Error summary mismatch: the notebook’s closing summary says crazing and rolled-in scale can be confused, but the detailed test output reports scratch -> inclusion. Use the detailed classification report and misclassification output when describing measured errors.

Demo split membership: the six-image demo selects images from the same directory tree and does not verify split membership. Do not call those images an independent test.

Training epochs: the training script default is 12 epochs, but saved training curves have different lengths. Do not state the exact run length without checking run logs.

Latency figures: the notebook reports Baseline 0.64 ms/image, ResNet-50 0.70, and EfficientNet-B0 0.55. A separate benchmark CSV in the project reports different Baseline and ResNet timings. Treat latency as run-specific and prioritize the notebook values only when discussing the notebook.

Augmentation and CLAHE: they are plausible design choices, but no ablation isolates their effect on test performance.

Grad-CAM: use "the visualization suggests attention near defect regions," not "the model understands the defect for this reason."

Formula reminders

Accuracy = correct predictions / all predictions. Precision = true positives / predicted positives. Recall = true positives / actual positives. F1 is the harmonic mean of precision and recall. Macro F1 is the unweighted average of per-class F1 scores.

Strong closing answer

"Transfer learning substantially outperformed the scratch-trained baseline on this NEU benchmark. EfficientNet-B0 matched ResNet-50 while using far fewer parameters, making it the more efficient choice in this comparison. The next milestone is external, group-aware validation on production data; the current holdout score alone does not establish manufacturing reliability."